THE HIDDEN ROUTING TAX OF TRANSFORMERS: FROM FORCED SERIALIZATION TO LATENT COGNITIVE DIFFUSION

A PREPRINT

Anonymous Researcher*
Independent Software Engineer
Full-Stack & Embedded Systems Developer
spore20ka@gmail.com

April 14, 2026

ABSTRACT

In the contemporary paradigm of Large Language Models (LLMs), the parameters within Attention and Mixture-of-Experts (MoE) layers are generally assumed to be dedicated entirely to semantic representation and logical reasoning. However, an analysis of the information physics within the classical residual stream reveals a critical bottleneck: the single-state routing mechanism. Deprived of a dedicated routing bus, stochastic gradient descent (SGD) repurposes functional parameters to establish heavy-weight communication protocols, imposing a severe "Hidden Routing Tax." This paper articulates how this architectural flaw leads to forced feature entanglement and hidden serialization. Furthermore, we reinterpret the phenomenon of PreNorm Dilution—often considered a mere mathematical defect—as an evolutionary adaptation, conceptualized as "Magnitude Growth as a Survival Strategy," wherein layers exponentially amplify their outputs to overcome interference in a blind pipeline. In MoE architectures, this unidirectionality induces "Defensive Duplication," compelling early experts to cache redundant primitives due to the absence of backward feedback loops. While recent advancements such as Attention Residuals (arXiv:2603.15031) provide a vital normalization of the pipeline, they remain a unidirectional compromise. To fully overcome these limitations, we propose the "Latent Cognitive Diffusion" architecture—a hybrid MoE-Diffusion framework that replaces the linear pipeline with an Iterative Consensus Loop, transitioning from unidirectional data transmission to multidimensional latent reflection.

Keywords Transformers · Mixture-of-Experts · Routing · PreNorm Dilution · Diffusion Models · Large Language Models

1 Introduction

The prevailing consensus in modern Large Language Model (LLM) architectures assumes that the billions of parameters constituting Attention blocks and Mixture-of-Experts (MoE) layers are exclusively allocated to productive cognitive tasks: storing factual knowledge, extracting patterns, and executing logical deductions. However, a rigorous examination of the physical transmission of information through the classical residual stream exposes a fundamental architectural vulnerability. Historically, the transmission pipeline along the depth axis has been left to its own devices. Devoid of a dedicated system routing bus, the optimization process—stochastic gradient descent (SGD)—has been forced to adapt. It autonomously repurposes highly valuable network parameters to synthesize computationally expensive communication protocols. In this paper, we deconstruct how this "Hidden Routing Tax" compromises the theoretical purity of expert layers, elucidate why vector magnitude inflation is actually an evolutionary survival strategy, and outline the impending paradigm shift toward diffusion-based cognitive architectures.

*This paper presents a conceptual framework and architectural hypothesis.

Epistemic Status: We state clearly that the following propositions constitute an architectural critique, a conceptual hypothesis, and a theoretical framework. This perspective is offered to re-evaluate the physics of information propagation in neural networks from an unorthodox vantage point. We do not claim this to be absolute empirical truth; the hypotheses presented herein demand rigorous empirical validation at a computational scale currently accessible only to the largest AI laboratories. Thus, this paper serves as an invitation to discussion—a provocation aimed at identifying profound architectural bottlenecks and charting the future trajectory of foundation models.

2 The Routing Tax & Forced Cryptography

The classical Transformer architecture (Vaswani et al., 2017) provides layers with only a single, shared vector-container (the residual stream) for data exchange. If an early layer (e.g., Layer 5) must transmit a critical factual representation to a consuming layer (e.g., Layer 50), this signal must survive transit through 44 intermediate blocks.

Because the network is left without explicit routing mechanisms, a hidden logistical infrastructure spontaneously emerges within the weights:

1. **Hidden Serialization (Packing):** The container is forced to memorize not merely the "knowledge vector" itself, but also the methodologies for encoding it. An early layer encrypts facts into a high-dimensional superposition, appending a latent key or address.
2. **Forced Feature Entanglement (Latent Serialization):** Intermediate layers expend their computational capacity to carefully transport these archived packets. They must entangle the transit signal with their own computational outputs, ensuring the payload is not destroyed by matrix multiplications.
3. **Deserialization (Unpacking):** Deeper layers must dedicate millions of parameters to function as decoders. They are obligated to filter accumulated noise, recognize the routing key, decompress the container, and only then proceed with the intended logical processing.

Ultimately, the model expends a monumental fraction of its "intelligence" functioning as a system router and ZIP-archiver, rather than performing pure inference.

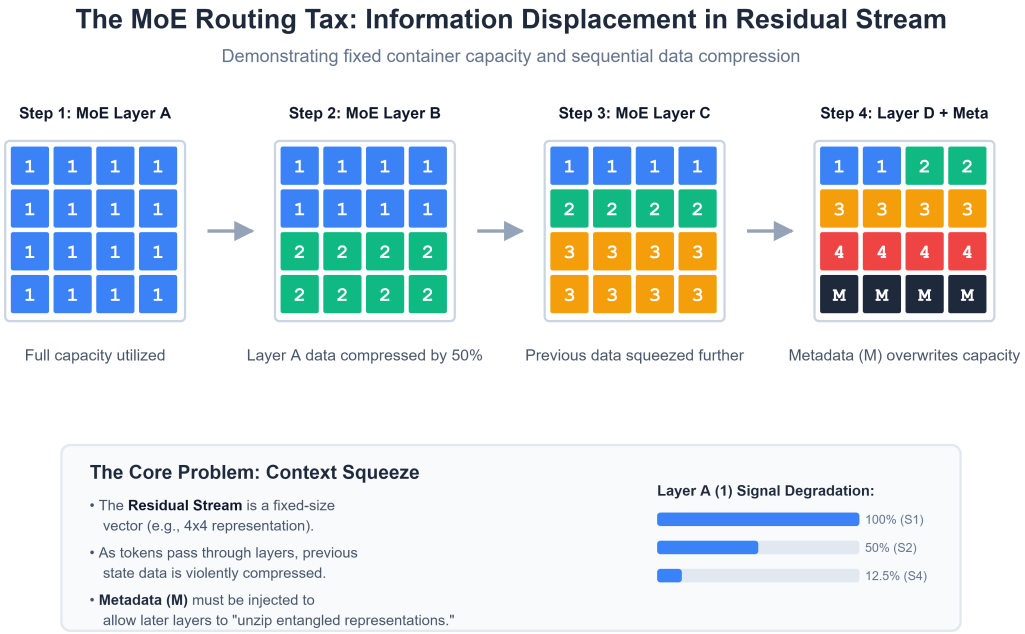

Figure 1: The Hidden Logistics Infrastructure of the Residual Stream. Intermediate layers function as forced routers, entangling and decrypting transit representations.

3 Magnitude Expansion as an Evolutionary Adaptation

Academic literature frequently documents the uncontrolled growth of vector values as they propagate through successive layers—a phenomenon known as PreNorm Dilution or Output Magnitude Growth. Traditionally, this is diagnosed as a purely mathematical flaw inherent to fixed accumulation in the residual stream.

Viewed through the lens of the routing bottleneck, however, the true physical cause becomes apparent: *Adaptive Magnitude Inflation* is a forced evolutionary adaptation.

When dozens of blocks attempt to write their encrypted archives into a single, narrow conduit, severe interference occurs. To guarantee its archive is not overwritten or diluted, an early layer resorts to mathematically "shouting." Layers are forced to exponentially increase the amplitude of their outputs so their signals dominate the noise of subsequent layers. Magnitude Growth as a Survival Strategy is a direct consequence of the network compensating for the absence of a proper routing bus.

4 The MoE Bottleneck & Defensive Duplication

The Mixture-of-Experts (MoE) paradigm (Shazeer et al., 2017) was conceived as an elegant repository of highly specialized knowledge. Yet, due to the compromised pipeline, these experts find themselves constrained. SGD forces them to be cryptographers as well as domain specialists.

Because of the unidirectional nature of the pipeline, a critical vulnerability emerges: the inability to revise, combined with a total lack of backward feedback loops. Early layers operate blindly; they cannot query terminal layers to ascertain which specific facets of data will be required at the conclusion of the logical chain.

Consequently, MoE routers engage in **Defensive Duplication**. They cache and transmit redundant primitives, fragmented contextual features, and overlapping information "just in case." It is precisely this architectural inability to iteratively revise intermediate representations that engenders parameter redundancy, leading to hallucinations and logical degradation during complex reasoning tasks.

5 Current Mitigations: The Attention Residuals Compromise

The industry is beginning to formally recognize the limitations of blind routing. A recent preprint by the Moonshot AI (Kimi) team, *Attention Residuals* (Kimi Team, 2026), introduces a profound architectural patch that acknowledges the constraints of modern computational resources.

Recognizing that a complete architectural overhaul is computationally prohibitive, the authors normalized the pipeline by replacing rigid accumulation with an Attention mechanism along the depth axis. Deeper layers acquired the capability to selectively query specific early layers, bypassing the need for intermediate compression. Consequently, the computational burden on the "containers" sharply decreased, and the effective data bus dimensionality expanded, allowing data retrieval with minimal compression.

While this is a remarkable achievement, it remains a compromise. The *Attention Residuals* architecture brilliantly solves the problem of retrieving from the past, but it leaves the pipeline strictly unidirectional. The initial MoE layers remain deprived of backward interaction with subsequent layers. Thus, the root cause of Defensive Duplication—the inability to iteratively revise hypotheses—remains unaddressed.

6 Vision: Latent Cognitive Diffusion

How can the architecture eliminate forced duplication and empower the network with true reflection? The conceptual answer lies in a hybrid of the MoE paradigm and the principles of diffusion models, operating directly within the latent space.

While implementing this currently demands virtually inaccessible computational resources, the generation process must ideally transition from a linear pipeline to an **Iterative Consensus Loop**:

1. **Primary Diffusion (Hypothesis Generation):** Diverse experts (MoE) parallelly decompress their independent knowledge slices, forming a multidimensional "cloud of raw hypotheses."
2. **Reverse Diffusion (Cross-Reflection):** All generated states do not proceed blindly down the pipeline. Instead, they are directed backwards and re-processed through each expert. A backward feedback loop emerges: a deep logic expert can correct and supplement the early data provided by a factual expert.

3. **Consensus Collapse:** This iterative denoising continues until the experts mutually resolve contradictions and eliminate noise. Only then does the hypothesis cloud collapse into a deterministic vector, entirely free of duplicated primitives.

Cycling data bi-directionally through expert pools implies a multifold decrease in inference speed and a massive increase in hardware requirements. Consequently, we currently navigate through compromises. Nevertheless, the trajectory is clear: future neural networks will evolve from unidirectional courier-archivers into *Latent Cognitive Diffusion* systems, where knowledge is not merely transmitted, but iteratively comprehended.

7 Conclusion & Future Directions

The identification of the Hidden Routing Tax exposes a fundamental inefficiency in current LLMs: the squandering of parameters on logistical routing rather than cognitive processing. While innovations like Attention Residuals mitigate the symptoms of PreNorm Dilution and forced serialization, they do not resolve the unidirectional blindness that causes Defensive Duplication. The theoretical shift toward Latent Cognitive Diffusion represents the next tectonic evolution. By replacing rigid directed acyclic graphs with Iterative Consensus Loops, models can reclaim the routing tax, transforming from vast ZIP-archivers of linguistic patterns into systems capable of deep, multidimensional reflection.

References

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, pages 5998–6008, 2017.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

Kimi Team (Moonshot AI). Attention Residuals. *arXiv preprint arXiv:2603.15031*, 2026.